Hongyi (Tom) Chen

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Data Science undergraduate spanning AI/DL research (medical LLMs · LLM theory), quantitative finance, and data analytics — driven by data-informed modeling and decision-making

Education

The Chinese University of Hong Kong, Shenzhen — **B.S. in Data Science**
Shenzhen, China — *Sep 2023 - Present*

- **GPA:** 3.5/4.0 (Top 25% of major)
- **Core Coursework:** Machine Learning, Optimization, Stochastic Processes, Discrete Mathematics, Ordinary Differential Equations, Probability & Statistics, Linear Algebra, Data Structures.
- **Honors:** Dean's List (2023–2024); Second Prize, Zhixing Social Practice Award.

University of California, Berkeley — **Visiting Student, Computer Science**
Berkeley, CA, USA — *Aug 2025 - Dec 2025*

- **Core Coursework:** Introduction to Artificial Intelligence, Computer Architecture, Deep Learning Model Design.

Professional Experience

China Merchants Bank (Xiamen) — **FinTech Intern**
Jul 2025 - Aug 2025

- **Model Optimization:** Optimized a text-classification model for anti-fraud detection; refined the training pipeline and tuned hyperparameters to improve discriminative power and generalization, raising AUC by 12% while reducing the false-positive rate.
- **Model Interpretability:** Built a token-level feature-attribution module enabling fine-grained prediction attribution and key-semantic identification, improving model-review efficiency.

DigitSuper Technology (Shenzhen) — **Data Analyst Intern**
Apr 2025 - Jun 2025

- **Feature Analysis:** Processed and analyzed advertising-placement data with Python and SQL, identifying the key drivers of conversion efficiency to inform placement strategy and product decisions; drove a 20% lift in core-product BSR.
- **Strategy Optimization:** Designed A/B testing frameworks for real-time-bidding (RTB) scenarios, validating and refining bidding strategies to improve ROAS by 10%.

Research & Projects

EHRFormer for Sudden Cardiac Death Prediction (Multi-Hospital Pretraining)
Advisor: Prof. Joshua Zhou — *Apr 2026 - Present*

- **Data Engineering:** Built a standardization pipeline for multi-hospital EHR data covering field alignment, missing-value imputation, time-series feature construction, and conversion to model-ready inputs; designed patient-level splits, leakage-prevention safeguards, and consistency checks to improve data reliability and evaluation credibility.
- **Model Fine-Tuning:** Applied LoRA for parameter-efficient fine-tuning of EHRFormer, achieving 85% reconstruction accuracy in the Phase 1 model; further planned self-supervised continued pretraining with masking strategies to strengthen representation capacity for the sudden-cardiac-death domain.
- **Downstream Application:** Contributed to downstream diagnostic-task and evaluation design, validating representation capacity and training-pipeline stability, and explored applications in reducing unnecessary examinations, predicting key clinical indicators, and supporting clinical risk assessment.

SkinGPT-X: Benchmarking Medical Large Language Models
Advisor: Prof. Joshua Zhou — *Dec 2025 - Present*

- **Benchmark Design:** Co-designed a systematic evaluation framework for state-of-the-art medical LLMs (e.g., Qwen3, MedGemma, HuLuMed).
- **Model Evaluation:** Assessed models across diagnostic accuracy and complex-case handling, validating the performance advantages of SkinGPT-X.
- **Error Analysis:** Conducted structured analysis of clinical-validation data to identify key failure modes on complex cases and guide model refinement.
- **Results:** Demonstrated an ∼8% performance gain across multiple metrics with greater stability and robustness on complex cases; findings submitted to *Nature Medicine* (Mar 2026, under review).

The Impact of Polysemanticity on LLM Model Editing GitHub Repo
Advisor: Prof. Anant Sahai *Nov 2025 - Dec 2025*

- **Experiment Design & Validation:** Independently designed and validated core experiments on how "polysemantic neurons" affect knowledge editing, showing that neuron-level editing (**FiNE**) induces more severe collateral damage than layer-level editing (**ROME**).
- **Framework & Analysis:** Built a complete validation framework on **Pythia-6.9B**, combining causal tracing and collision analysis to localize polysemantic neurons, quantify collateral damage, and characterize the precision–robustness trade-off between FiNE and ROME.

Regime-Adaptive Stock Trend Prediction with GRU GitHub Repo
Course Project (CUHK-Shenzhen) *Nov 2025 - Dec 2025*

- **Architecture Optimization:** Built a lightweight GRU time-series model with LayerNorm and Tanh activations to improve training stability on low-SNR financial data, mitigating vanishing gradients and non-convergence.
- **Market-Regime Modeling:** Constructed a balanced, outlier-trimmed training set (excluding the 2020 extreme regime) to reduce bull-market bias and improve out-of-sample adaptability.
- **Objective Alignment:** Replaced MSE loss with the Information Coefficient (IC) to align training with a Top-K stock-selection strategy, yielding stronger prediction–strategy consistency and superior risk-adjusted returns in backtesting.

Ranno.ai: On-Device AI Agent for Route Planning (Google AI Hackathon) GitHub Repo
Full-Stack Developer *Oct 2025*

- **Agent Intent Parsing & Tool Calling:** Built an on-device NLU module with the Chrome Prompt API / Gemini Nano to parse natural-language requests into structured task intents, then invoked the Google Maps API (routes, places, geometry) as tools to fulfill them.
- **Multi-Source Data Fusion:** Further fused San Francisco crime-rate and urban-amenity data to build a safety-aware hybrid route-planning system.
- **System Implementation:** Developed the front-end with React (Vite) and TypeScript, and implemented back-end path-generation logic.

Leadership & Extracurricular

Zhixing Social Practice Project
Project Lead *Jun 2024 - Oct 2024*

- Led data processing and analysis, using Python (Pandas) for survey-data cleaning and feature construction, and applied K-Means clustering to feedback to extract four core user personas.
- Delivered a data-analysis report whose key findings were adopted by CUHK-Shenzhen to optimize student-affairs strategy.

Rural Revitalization Field Program
Co-Lead *Aug 2024*

- Co-led multi-stakeholder research design and execution, directing field interviews and primary-data collection with government officials and villagers.
- Structured interview content into text data and applied thematic analysis to extract key challenges and policy feedback on rural industrial development.
- Authored a research report adopted by the local government and featured by regional media.

Skills

Programming & Web	Python, R, SQL, Java, C, JavaScript, HTML/CSS, LaTeX, Git
ML & Data	PyTorch, Scikit-learn, Transformers, Pandas, NumPy, Tableau
Languages	Mandarin (native), English (TOEFL 109, professional working proficiency)